

Safe Exploitation Without a Game Value: Restricted Responses in Three-Player Leduc Hold'em

Draft v9. Supersedes v8. All sections now on the 5,000-iteration baseline; worst cases use a pessimistic punisher throughout; solution stability and the second equilibrium are confirmed by converged runs.

Tags: **[EXACT]** exact game-tree evaluation on the 5,000-iteration baseline. **[MC]** Monte Carlo. All payoffs are chips per hand, measured relative to the seat's own equilibrium value unless stated otherwise.

Abstract

In two-player zero-sum games, safe opponent exploitation is well understood: restricted Nash responses trace the frontier between profiting from an opponent's mistakes and becoming exploitable, and safety is measured against the game's value. Neither tool carries over to three or more players, where there is no unique game value and an equilibrium strategy guarantees nothing. We study the problem exactly in three-player Leduc Hold'em. Against opponents built by biasing a converged equilibrium toward raising, calling or folding, we find: (i) equilibrium play is not a floor — a fixed equilibrium strategy loses up to 0.69 chips/hand to extreme opponents, an almost-indifferent opponent can shift up to 0.23 chips/hand between the other two players, and two self-interested, non-colluding opponents who keep learning converge, from every start we tried, to an equilibrium that costs a fixed equilibrium player 0.39 chips/hand; (ii) a three-player restricted Nash response that trusts neither opponent's model achieves **safe exploitation** — at least 0.05 chips/hand more profit with no material loss of worst-case safety — in 41 of 45 settings at moderate bias (median extra profit 0.33), 9 of 12 at intermediate bias and 8 of 15 at mild bias; (iii) it dominates naive mixing of equilibrium and best response in 162 of 166 comparisons at equal profit; and (iv) worst-case safety is far more sensitive than profit to small strategy differences and to how a near-indifferent punisher breaks ties, which shapes how any such comparison must be made.

1. Introduction

An equilibrium strategy cannot be exploited but leaves money on the table against weak opponents; a best response wins the most against a specific opponent but can be exploited in turn. In two-player zero-sum games this tradeoff has a well-developed theory. Restricted Nash responses (Johanson, Zinkevich & Bowling, 2007) trace its Pareto frontier, epsilon-safe strategies bound worst-case loss (McCracken & Bowling, 2004), and safe exploitation risks only what has already been won relative to the game value (Ganzfried & Sandholm, 2015).

Every one of those guarantees rests on the minimax theorem. With three players there is no game value to be safe relative to, an equilibrium strategy guarantees nothing, and one opponent's mistake need not become your gain. The tradeoff is therefore not just unmeasured in multiplayer poker but not yet well defined. Measuring it requires a game small enough for exact best responses and exploitability, and large enough that three players genuinely interact.

Three-player Leduc Hold'em is that game. Our contributions:

1. **Equilibrium is not a floor.** Concrete cases where a converged equilibrium player loses to deviating opponents, a measurement of where their mistakes' value goes, and a second equilibrium — reached by independently learning opponents — that costs a fixed equilibrium player 0.39 chips/hand.
2. **A three-player restricted Nash response** with symmetric, self-interested restriction, and an exact evaluation across 15 ordered opponent pairs, three seats and two bias levels.
3. **A sensitivity analysis of the worst-case measure.** Worst-case safety moves by up to 0.1 chips/hand under strategy changes that leave profit unchanged; we calibrate comparisons to this noise and to punisher tie-breaking.
4. **Structure of the safe region:** where it lies, which opponent is dangerous to trust, and how robust it is to modelling errors.

2. Related work

Robust and safe exploitation. Restricted Nash responses pin the opponent to a model with probability p and otherwise let it best-respond, tracing a Pareto-optimal gain/exploitability frontier (Johanson et al., 2007); data-biased responses weight the pinning by data confidence (Johanson & Bowling, 2009). Epsilon-safe strategies (McCracken & Bowling, 2004) and safe opponent exploitation (Ganzfried & Sandholm, 2015) bound risk relative to the game value. Li & Huang (2026) schedule the restriction from confidence sequences with certified exploitability in two-player Leduc. All are two-player zero-sum. Ganzfried's safe equilibrium

(2023) generalises restricted responses to multiplayer and non-zero-sum games and is the closest theoretical relative of our construction.

Opponent modelling. Leduc Hold'em was introduced with Bayesian opponent modelling (Southey et al., 2005); short-horizon exploitation in Kuhn poker was studied by Hoehn et al. (2005) and Southey, Hoehn & Holte (2009). Our identifier is Bayesian reasoning over hypothesised types (Albrecht, Crandall & Ramamoorthy, 2016), including parameterised types (Albrecht & Stone, 2017; survey in Albrecht & Stone, 2018). The only multiplayer precedent we found is Ganzfried, Wang & Chiswick (2024), who model opponents in three-player Kuhn poker. Showdown data are missing not at random (Guo, 2026); our likelihood models fold decisions and so handles this in principle.

Multiplayer equilibrium. CFR (Zinkevich et al., 2008) has no Nash guarantee beyond two players, though it has produced strong three-player (Abou Risk & Szafron, 2010) and superhuman six-player agents (Brown & Sandholm, 2019). Three-player Kuhn poker already has a parameterised family of equilibria (Szafron, Gibson & Sturtevant, 2013), which foreshadows the equilibrium selection effects we measure. Exact multiplayer methods now exist for small games (Ganzfried, 2026). The observation that an opponent's mistake can benefit a third player is known to poker players as Morton's theorem or implicit collusion; we quantify it against an equilibrium baseline.

3. Setup

3.1 Game

Three-player Leduc Hold'em in OpenSpiel (Lanctot et al., 2019). The deck scales with player count as $2 \text{ suits} \times (\text{players} + 1) \text{ ranks}$, so the three-player game uses 8 cards across 4 ranks. It has 25,800 information sets and 1,831,601 game-tree nodes, small enough for exact evaluation.

3.2 Equilibrium baseline [EXACT]

CFR+ for 5,000 iterations; `nash_conv` 0.0023, with each seat's unilateral best-response gain below 0.001. Equilibrium values by seat: **-0.0949, -0.0089, +0.1038** (seat 0 acts first). Position is worth about 0.2 chips/hand from first to last actor.

3.3 Opponents

Each opponent tilts the equilibrium toward one action, wherever that action is legal:

□

$\sigma'(a | I) = (1 - \alpha) \sigma(a | I) + \alpha [a = \text{favoured}]$

Types: **O** equilibrium, **R** over-raiser, **C** over-caller, **F** over-folder. Pairs are written in seat order of the two opponents (for a learner in seat 0, **FR** means seat 1 folds too much and seat 2 raises too much); all 16 ordered pairs are used, with **OO** as a control.

alpha is an override rate where the favoured action is legal, so it is not comparable across types: raising is illegal at the cap and folding is illegal with no bet to face. Per unit of alpha, an over-raiser deviates on 1.41 decisions per hand, an over-caller on 0.89, an over-folder on only 0.28.

3.4 Metrics

- **Gain:** our expected payoff against the biased pair, minus our seat's equilibrium value.
- **Worst case:** our expected payoff if *either* opponent abandons its model and plays the exact best response to our strategy, maximising its *own* payoff, while the other keeps playing its model; we take the worse of the two, minus our equilibrium value. The switching opponent is omniscient about our strategy, so this bounds a realistic punisher. An adversarial variant (minimising our payoff) also measures spite and is reported only as a bound.
- **Baseline:** the best of three equilibrium-level strategies, on each axis separately — our 5,000-iteration CFR table, the same run at 1,000 iterations, and the $p = 0$ restricted response (§4.5).
- **Safe exploitation:** some $p > 0$ earns at least **0.05** more than the baseline, with a worst case no more than **0.05** below the baseline's.

Worst case, pessimistically. When a punisher is nearly indifferent between responses, which one an exact best-response computation happens to pick can change our payoff substantially (§4.3). All worst cases in §7 therefore use a *pessimistic* punisher: it maximises its own payoff, but among responses within 0.01 chips of its best at each decision it picks the one worst for us.

Noise scale. Every number is an exact evaluation of the strategy computed, but strategies are approximate solutions. Re-solving six restricted responses with twice the iterations, or from a random start, changed gain by at most 0.010 (median 0.001) and worst case by at most 0.030 (median 0.007). The 0.05 tolerances above are therefore conservative.

3.5 Methods

- **Linear mixing:** $(1 - w) \cdot \text{CFR} + w \cdot \text{BR}(\text{model})$, $w = 0, 0.1, \dots, 1$.
- **Three-player restricted Nash response (RNR).** In a modified game each opponent, independently, follows its biased model with probability p and is otherwise *free*: it knows its mode and plays to maximise its own payoff. We observe neither mode. Solved with CFR+ (600 iterations; largest convergence gap across all runs 0.0076). At $p = 0$ both opponents are always free, so the modified game *is* the real game and RNR is an independent, shorter CFR+ solve of it (600 iterations; players' combined best-response gain 0.0096, against 0.0023 for the CFR table). Making both free opponents minimise our payoff would make them a coalition; we

exclude that. Restricting only one opponent (a one-sided variant) was studied and rejected: the trusted opponent can punish it for over 3 chips/hand (§7.3).

- **Engine.** A vectorised tree engine for CFR on modified games, best responses and exact evaluation, verified against OpenSpiel (§8).

4. Equilibrium is not a floor

4.1 Extreme opponents beat a non-adapting equilibrium player [EXACT]. Playing CFR against two over-raisers at $\alpha = 1.0$ loses 0.185 chips/hand relative to equilibrium; against an over-raiser and an over-caller, 0.692. In two-player zero-sum games, deviations can only help an equilibrium player. Here the equilibrium carries no floor. This value doubled when the baseline converged from 1,000 to 5,000 iterations: at extreme bias, play is driven into rarely visited parts of the tree where equilibria are least pinned down.

4.2 Opponents' mistakes go to the third player [EXACT]. With one over-raiser in seat 1, an equilibrium opponent in seat 2, and us playing CFR in seat 0, the raiser loses 0.329 per hand at $\alpha 0.35$; we gain 0.126 and the other opponent gains 0.203. At $\alpha 0.1$ the split is -0.086 , $+0.023$, $+0.063$. At small and moderate bias the third player collects more of the mistake than we do — a quantitative Morton's theorem. Adapting is how you claim it.

4.3 An almost-indifferent opponent is a kingmaker [EXACT]. With all three at equilibrium, an opponent that best-responds to the others gains almost nothing, so it is nearly indifferent among many responses. Restricted to responses within 0.001 chips of its best at each decision, it can still move *our* payoff over a wide range: from seat 1 with seat 2 deviating, from -0.23 to $+0.11$; from seat 0 with seat 2 deviating, from -0.11 to $+0.23$. An opponent with nothing to gain can decide, at almost no cost to itself, which of the other two players pays. (An earlier draft reported single values for these effects; they reflected tie-breaking in the best-response computation, not a property of the game.)

4.4 Learning opponents reach a second equilibrium [EXACT]. Holding our CFR strategy fixed in seat 0, the two opponents learn simultaneously, each self-interested, with no collusion. From a uniform start and two random starts, all three runs converge to the same profile (best-response gaps below 0.0012, tighter than our CFR baseline's own convergence): **we lose 0.391 while seat 1 gains 0.141 and seat 2 gains 0.251** relative to equilibrium. The equilibrium our CFR table belongs to is reached from none of them. Both opponents prefer the new equilibrium, and neither needs to coordinate to reach it. **A fixed equilibrium strategy is not safe against opponents who keep learning.**

4.5 Worst case is sensitive to which equilibrium-level strategy you play [EXACT]. Three approximate equilibria of the same game — our 5,000-iteration CFR table, the same OpenSpiel

run stopped at 1,000 iterations, and the $p = 0$ restricted response — earn nearly the same when everyone plays equilibrium, yet differ markedly in worst case (seat 0, alpha 0.35):

pair CFR, 5,000 it. CFR, 1,000 it. RNR $p = 0$

FO	-0.025	+0.067	+0.070
RO	-0.103	-0.071	-0.050
RR	-0.194	-0.174	-0.132

The two CFR checkpoints differ in strategy by only 0.004 (reach-weighted total variation), so the worst-case measure is highly sensitive to small strategy differences. The $p = 0$ solve differs mostly at rarely reached information sets (0.080 there, 0.008 where play usually goes), which equilibrium conditions barely constrain but a punisher's deviation reaches. In every pair tested, the more converged CFR checkpoint was the more punishable. The pattern does not extend to restricted responses, whose worst cases moved by at most 0.03, in either direction, when re-solved with twice the iterations (§8). Why the CFR checkpoints differ we have not established. We therefore measure adaptation against the **best of all three** on each axis; the headline in §7 is unchanged whether or not the 1,000-iteration strategy is included.

5. The value of exploiting mistakes

Seat 0, alpha 0.35 **[EXACT]**: gain available to a best response, and what CFR collects without adapting:

pair CFR gain BR gain pair CFR gain BR gain

FF	+0.171	+1.846	RR	+0.219	+1.217
RF	+0.207	+1.415	CC	+0.189	+0.687
FR	+0.326	+1.379	OF	+0.088	+0.950
CF	+0.199	+1.191	RO	+0.126	+0.879
CR	+0.296	+1.087	OR	+0.240	+0.802
RC	+0.208	+0.997	OC	+0.105	+0.489
FC	+0.183	+0.972	CO	+0.095	+0.418
FO	+0.077	+0.776			

Best responses are catastrophically exposed: their worst case ranges from -1.1 to -4.0 .

Shape and scale [EXACT]. Best-response gain grows monotonically with alpha, near linearly for $\alpha \leq 0.2$. Compared at equal *deviation* rather than equal alpha, folding errors are by far the most punishable: at 0.10 overridden decisions per hand, one over-folder is worth +0.98 against +0.10 for an over-raiser and +0.16 for an over-caller. Apparent saturation of over-folder gains at high alpha is an artifact of alpha running out of room, not of folding errors becoming harmless.

6. Identifying opponents [EXACT likelihoods, MC sessions]

An exact Bayesian posterior over the 16 ordered pair hypotheses, marginalising over unknown hole cards, needs no privileged information. Hands until the posterior on the true pair reaches 0.9 (seat 0, 50 sessions per pair; median over pairs of each pair's median):

alpha actions only + showdown cards + all cards revealed

0.10	112	108	54
0.20	46	45	31
0.35	22	19	14

Revealing cards after every hand helps 1.4–2.1×; showdown cards alone barely help, because the betting already carries most of the information. At alpha 0.1 some pairs are not identified within 250 hands in up to 60% of sessions. Rare actions move the belief most: where the equilibrium plays (raise, call, fold) = (0.4, 0.1, 0.5), a call is 4.15× more likely from an over-caller at alpha 0.35, a raise only 1.53× more likely from an over-raiser. Classifying a single opponent is 2–4× faster than pinning the ordered pair **[1,000-iteration baseline; not rerun]**.

The deviation most valuable to exploit (folding, §5) is also among the slowest to detect: 45 hands for FO versus 31 for RO at alpha 0.35, and 7–9 for pairs of two deviators such as RR and RC.

7. Restricted Nash response

7.1 Headline [EXACT, pessimistic punisher]

Safe exploitation (≥ 0.05 extra profit, worst case within 0.05 of baseline)	41 of 45 at alpha 0.35; 9 of 12 at alpha 0.2 †; 8 of 15 at alpha 0.1
... median extra profit where it exists	+0.33 at alpha 0.35 (39% of a best response's extra); +0.16 at 0.2; +0.09 at 0.1
... requiring ≥ 0.10 extra profit	39 of 45 at alpha 0.35; 2 of 15 at alpha 0.1
RNR vs linear mixing at equal gain	better in 162 of 166 points; median +0.69 of worst case

† alpha 0.2: seat 0 only, 12 of 15 pairs completed, exact rather than pessimistic punisher.

A stricter question — is the restricted response *both* more profitable *and* at least 0.05 safer than the baseline? — holds in 33 of 45 settings at alpha 0.35 and 1 of 15 at alpha 0.1. We use safe exploitation as the headline, since earning more without becoming less safe is what a restricted response is for.

7.2 Seat 0, alpha 0.35 [EXACT, pessimistic punisher]

Baseline (best of the three equilibrium-level strategies on each axis) and the most profitable safe restricted response (gain / worst case):

pair	baseline	best safe RNR	best response
RR	+0.231 / -0.135	p=0.75: +0.954 / +0.063	+1.217 / -1.107
RF	+0.215 / -0.058	p=0.5: +0.807 / -0.007	+1.415 / -3.320
RC	+0.217 / -0.050	p=0.5: +0.675 / +0.079	+0.997 / -1.200
CC	+0.192 / -0.126	p=0.9: +0.621 / -0.036	+0.687 / -1.962
FF	+0.171 / -0.046	p=0.5: +0.575 / +0.039	+1.846 / -3.860
CR	+0.306 / -0.135	p=0.5: +0.653 / +0.043	+1.087 / -1.163
FR	+0.328 / -0.135	p=0.5: +0.661 / +0.047	+1.379 / -2.519
CF	+0.199 / -0.126	p=0.5: +0.481 / -0.010	+1.191 / -3.726
FC	+0.185 / +0.035	p=0.5: +0.439 / +0.063	+0.972 / -3.969
CO	+0.095 / -0.126	p=0.9: +0.338 / -0.097	+0.418 / -2.211
OF	+0.088 / -0.057	p=0.5: +0.311 / -0.008	+0.950 / -3.419
RO	+0.129 / -0.050	p=0.25: +0.350 / -0.063	+0.879 / -1.178
OC	+0.106 / +0.035	p=0.5: +0.293 / +0.042	+0.489 / -2.319
OR	+0.242 / -0.135	p=0.5: +0.375 / +0.124	+0.802 / -1.709
FO	+0.077 / -0.024	p=0.25: +0.141 / -0.047	+0.776 / -3.104

Typically the safe restricted response two- to four-folds the baseline's profit while its worst case stays close to the baseline's, often above it; a best response's worst case is -1.1 to -4.0.

7.3 The safe region, and its abrupt end [EXACT]

FF, seat 0, fine grid in p:

p	0	0.25	0.5	0.6	0.65	0.7	0.75	0.9	0.97
gain	+0.170	+0.299	+0.575	+0.974	+1.323	+1.439	+1.580	+1.708	+1.796
worst	-0.056	+0.123	+0.039	-0.339	-0.521	-0.572	-0.528	-0.531	-1.309

The computed solutions show two regimes: a *safe* regime up to $p = 0.5$ where the worst case stays above equilibrium, and an *exploit* regime from $p = 0.65$ where exposure plateaus near -0.53 while gain keeps rising, then a final cliff near the pure best response. The switch is abrupt in the solutions we computed. Because the modified game is general-sum and CFR+ carries no uniqueness guarantee there, we cannot rule out that the jump reflects the solver moving between different solutions of the modified game rather than a property of restricted responses as such. RO shows the same shape from seat 2, with the switch between $p = 0.25$ ($+0.588 / +0.141$, against CFR's $+0.212 / -0.088$) and $p = 0.5$ ($+0.763 / -0.289$).

One-sided restriction fails. Trusting one opponent's model completely gives gains of $+1.28$ to $+1.69$, but that opponent can then punish for -3.2 to -3.6 . Draft v5 of this work reported the one-sided variant as a success because the worst case was only checked against the untrusted opponent. The symmetric version at $p = 0.9$ achieves a similar gain ($+1.708$) with a worst case of -0.531 .

7.4 By seat [EXACT, pessimistic punisher]

Safe exploitation at alpha 0.35:

seat settings median extra profit

0	15 of 15	+0.282
1	14 of 15	+0.324
2	12 of 15	+0.401

There is no clear seat ordering in how often safe exploitation exists once each seat is measured against its own equilibrium value. The strongest single point: RR from seat 2 at $p = 0.5$, $+1.016$ gain with a worst case of $+0.453$ to $+0.475$ across three independent solves.

7.5 Weaker bias [EXACT]

Safe exploitation exists in 9 of 12 completed pairs at alpha 0.2 (median extra $+0.16$) and 8 of 15 at alpha 0.1 (median extra $+0.09$; only 2 offer $+0.10$ or more). FF, seat 0, alpha 0.1 (pessimistic punisher):

	CFR	$p=0.25$	$p=0.5$	$p=0.75$	$p=0.9$	BR
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gain	$+0.049$	$+0.057$	$+0.083$	$+0.143$	$+0.314$	$+0.379$
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worst	-0.066	-0.034	$+0.036$	$+0.001$	-0.527	-3.536
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Mild deviators are both slower to identify (§6) and less valuable to exploit safely.

7.6 Which opponent is dangerous to trust [EXACT]

Split by which opponent switches to punishing (seat 0, alpha 0.35, relative to equilibrium):

pair p seat 1 punishes seat 2 punishes

OF 0.9 +0.547 **-0.834** (the folder)

RO 0.9 **-0.498** (the raiser) +0.389

OR 0.9 +0.172 **-0.709** (the raiser)

Swapping the raiser's seat (RO vs OR) moves the danger with him. Across all seats and alphas, when one opponent deviates and the other plays equilibrium, **the deviator is the harder punisher in 56 of 68 cases**, and in 49 of the 54 cases where the two punishers differ by at least 0.10. A natural interpretation — our commitment to their pattern is what they can turn against us — is a hypothesis we have not tested directly. Exceptions are mostly over-caller pairs, where commitment is smallest (e.g. OC at $p = 0.9$: the equilibrium opponent punishes for -0.040, the over-caller not at all, +0.224). With two deviators there is no simple rule.

7.7 Robustness to model errors [EXACT]

Strategies tuned for FF at alpha 0.35 (seat 0), played against other opponents (gain):

	FF 0.1	FF 0.2	FF 0.35	FF 0.5	RR	CC	OO
CFR	+0.049	+0.098	+0.171	+0.242	+0.219	+0.189	0.000
RNR 0.25	+0.082	+0.173	+0.299	+0.410	+0.179	+0.209	-0.015
RNR 0.5	+0.131	+0.319	+0.575	+0.797	+0.187	+0.117	-0.072
RNR 0.75	+0.118	+0.749	+1.580	+2.258	-0.069	-0.313	-0.569
RNR 0.9	+0.017	+0.750	+1.708	+2.475	-0.192	-0.530	-0.785
BR	-0.027	+0.789	+1.846	+2.677	-0.933	-0.961	-0.920

At $p \leq 0.5$ the strategy is robust: the right type with the wrong alpha still beats CFR, and the wrong type costs at most 0.07. At $p \geq 0.75$ the wrong type costs 0.3–0.8. For this pair, the regime switch of §7.3 is also where robustness to modelling error collapses. That suggests a rule for combining identification with adaptation — stay at low p until the posterior is

confident — but it rests on one pair and one seat and should be tested more broadly before being stated as general.

7.8 Where safe exploitation is not found

At alpha 0.35: FO, OC and OF from seat 2, and RO from seat 1. At alpha 0.2: OR, RO, FO (seat 0). At alpha 0.1: CF, CO, FO, OF, OR, RO, RR. Nearly all are single-deviator pairs, where there is less to gain and the baseline already offers comparable protection. Several were tested at only a few values of p ; where a safe region exists it usually lies at $p \leq 0.5$ and can end below $p = 0.5$.

7.9 When both opponents adapt [EXACT]

If both opponents learn best responses to our fixed strategy and to each other (500 iterations, gaps below 0.0012), exposure is far larger than against a single punisher (FF, seat 0):

single punisher (pessimistic) both adapt		
CFR	-0.061	-0.391
RNR $p = 0.5 + 0.039$		-0.213
RNR $p = 0.9 - 0.531$		-2.246
BR	-3.860	-4.289 ‡

‡ 120 iterations, gaps up to 0.011.

RNR at $p = 0.5$ is hurt roughly half as much as CFR. These are static stress tests: in practice we would observe the opponents' change and adapt back, so the relevant cost is exposure multiplied by the time to detect the change (§6).

8. Validity checks

check	result
Engine vs OpenSpiel	identical to 6 decimals for expected values and best responses, including a best response against a biased, non-equilibrium profile
All results on the same baseline	saved strategies from every stage re-evaluate with 0.0 difference on the 5k baseline
$p = 0$ reuse	cached and from-scratch results identical to every digit

check	result
Solution stability	six settings re-solved with 1,200 iterations and from a random start: gain changes ≤ 0.010 , worst case ≤ 0.030
Pessimistic punisher	applied to all 64 settings; median change in any worst case 0.000, but up to -0.24 in some, so it is the default
Tie-breaking, all at equilibrium	large: our payoff can range over ± 0.23 (§4.3); single-value claims withdrawn
Second equilibrium	three starting points converge to the same profile, gaps < 0.0012
Baseline choice	safe exploitation measured against the best of three equilibrium-level strategies
Deviator claim	holds with a 0.10 materiality threshold (49 of 54)
OO control	no gain to find, as expected

9. Discussion

Safety needs a new reference point. Two-player safe exploitation measures risk against the game value. In three-player games there is no such value, a single equilibrium is not a floor, and learning opponents can move to an equilibrium that is worse for you. Our results suggest a practical substitute: measure risk against the best available equilibrium, under a self-interested — not spiteful — punisher.

How you adapt matters more than how fast. Mixing equilibrium with a best response is dominated at nearly every point. A restricted response that trusts neither opponent fully finds a safe region, usually at low trust, where profit rises several-fold without a material loss of robustness.

Compare safety carefully. Profit is a smooth function of strategy; worst-case safety is not. Two strategies that earn the same can differ by 0.1 chips/hand in how punishable they are, and when every player is at equilibrium the punisher's choice among near-equal responses matters as much as its best response. Safety comparisons in multiplayer games need a stated noise scale.

Most of the risk comes from the opponent you are exploiting. A policy committed to an opponent's mistake is exposed to that opponent's correction. This suggests restricting more tightly against the deviator than against the other player.

10. Limitations and open items

1. **Opponent class:** single-action tilts, equal alpha for both opponents. Heterogeneous pairs and richer styles are untested.
2. **Persistent types:** RNR redraws opponent modes every hand; real opponents keep their style.
3. **Identification and adaptation are not yet combined:** choosing p from the posterior is the natural next experiment, guided by §7.7.
4. **Per-opponent restriction** is untested; §7.6 suggests restricting more tightly against the deviator.
5. **Choice of equilibrium-level baseline.** Worst cases differ across equilibrium-level strategies (§4.5). A refinement that pins down rarely reached play (e.g. a trembling-hand style perturbation) is the principled fix; exact multiplayer solvers (Ganzfried, 2026) could enumerate alternatives.
6. **alpha 0.2** covers seat 0 and 12 of 15 pairs, with the exact rather than pessimistic punisher; alpha grid otherwise 0.1 and 0.35.
7. **Mechanisms untested:** the explanation offered for §7.6 (commitment to a deviator's pattern) is a hypothesis; §4.5's CFR-checkpoint difference is unexplained.
8. **Single-opponent identification speed** (§6) is from the 1,000-iteration baseline.

11. Conclusion

In three-player Leduc Hold'em, equilibrium play offers no floor: extreme opponents beat it, an almost-indifferent opponent can decide who pays, and independently learning opponents can settle into an equilibrium that costs a fixed equilibrium player 0.39 chips/hand. Yet safe exploitation is possible: a symmetric three-player restricted Nash response earns substantially more than any equilibrium-level strategy we tested without a material loss of worst-case safety in 41 of 45 settings at moderate bias, and dominates naive mixing almost everywhere. Worst-case safety is far more sensitive than profit to small changes in strategy, a fact any comparison of multiplayer strategies has to account for.

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✓ checked during drafting; Δ confirm before submission.

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Appendix: reproduction

`rn timer package/`: `tree_engine.py` (build and verify the engine), `rn timer general.py` (restricted responses and linear baselines), `run_queue.py` (all jobs), `summarize.py`, `seed_p0_cache.py`, `gaps_check.py` (misspecification and both-adapt). Earlier analyses: `alpha_study/`. Results: 241 restricted-response solves and 64 linear baselines, all on the 5,000-iteration CFR+ solution.